Judge's Review & 10/10 Improvement Plan

Mekong Smart Land System (Project B) - Asian Hackathon for Green Future 2026, Track 3: Water Resources and Climate-Resilient Agriculture. Written from the judge's chair: every mistake found, why it costs points, and exactly how to fix it.

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1. Executive Summary - The Judge's Verdict

Project B is the correct base: it is a real, working, deployable prototype whose concept (multi-stakeholder climate decision support with crop-switch economics) matches the Track 3 brief almost line for line. It is currently a 5/10 product. It is held back by exactly three classes of problems: **(1) fabricated data and metrics** that will be exposed by this specific judging panel, **(2) serious security failures**, and **(3) scope sprawl** that dilutes a 5-minute demo. None of these is hard to fix. The plan in this document is ordered so that the highest-scoring fixes come first.

A word of honesty about "10/10 in all areas": some tens are reachable tonight (presentation, credibility, deployment safety); others (a validated ML model, measured real-world impact) are only reachable as a credible roadmap. Judges do not expect a student MVP to be a finished product - they expect it to be **honest, working, and defensible**, with a believable path to the rest. That combination is what actually scores 10 at a hackathon, and it is fully achievable with the fixes below. Every fix is tagged: **NOW** (do in the next hours), **BEFORE DEMO** (before you present), or **ROADMAP** (present as future work - saying it honestly earns points).

Criterion	Now	After NOW fixes	Ceiling with roadmap	Main blocker today
Innovation	6	9	10	Key innovation (switch economics) buried among 19 pages
Technical Depth	4	7	10	No real model; "AI/LSTM" labels with nothing behind them
Engineering Quality	5	8	10	Duplicated scoring logic, zero tests, template README
Scientific Credibility	3	9	10	Synthetic sine-wave data + fabricated back-test numbers
Presentation Quality	6	9	10	No story arc; navigation sprawl
Business Potential	5	8	10	No named first customer, no pricing logic
Practical Deployment	5	9	10	Leaked admin key; open write endpoint
Impact	4	8	10	Impact numbers computed over fabricated data
Scalability	5	8	10	Hardcoded 26 districts, no ingestion pipeline

Official scorecard mapping: Scientific Credibility + Technical Depth + Engineering + Deployment feed **Product Feasibility & Quality (40%)**; Impact + Scalability feed **Social & Environmental Impact (30%)**; Innovation feeds **Creativity (15%)**; Presentation + your Q&A conduct feed **Presentation & Critical Thinking (15%)**. Fixing credibility is therefore the single highest-leverage move: it sits inside the 40% bucket AND removes your biggest Q&A vulnerability.

2. Critical Issues - Fix Before Anything Else

These three items are not point-losers; they are disqualifiers. Everything else in this document is optimization. These are survival.

C1. Leaked Supabase service-role (admin) key	NOW
The mistake	vercel.json commits the SUPABASE_SERVICE_ROLE_KEY in plaintext. This key grants full admin access to your database and it is inside a zip file that has already been shared. The key must be treated as compromised.
Why it costs points	If any judge or mentor glances at the repo, this is an instant credibility kill for a "government-grade" platform. Worse, someone could wipe or poison your demo database DURING the event.
The fix	Rotate the key in the Supabase dashboard immediately (Settings -> API -> regenerate). Remove all secrets from vercel.json; set them as Vercel environment variables instead. The frontend should only ever see the anon key. Total time: 15 minutes.

C2. Fabricated back-test results on a real disaster	NOW
The mistake	The seed data hardcodes ai_prediction_accuracy: 0.82 and avoided_losses_usd: $120,000,000 for the real 2016 Mekong salinity crisis (and 0.88 / $185M for 2020). No model produced these numbers - they are invented and attached to a real tragedy that affected real people.
Why it costs points	Two judges (Assoc. Prof. Nguyen Trung Thanh, data science; Prof. Duong Nguyen Vu, decision-support systems) will ask "how was this accuracy computed?" There is no answer. A third (Dr. Nguyen Duy Dat) is a Vietnamese environmental engineer who knows these crises personally. Being caught fabricating results about the 2016 disaster, in Vietnam, in front of Vietnamese judges, ends your competition.
The fix	Option A (best, ~3-4 hours): compute a real, simple back-test. Download CHIRPS rainfall for 2015-2016 over the delta, compute a rainfall-deficit + trend index, show the date your threshold trips vs. the documented crisis onset - report the real lead time, whatever it is. Option B (acceptable, 20 minutes): relabel the panel "Illustrative mock-up - methodology in development" and remove the two fake numbers entirely. Never present them as results.

C3. Unauthenticated write access to the database	NOW
The mistake	api/recommendations.js accepts POST requests and inserts req.body directly into the database using the service-role client (which bypasses Row-Level Security). CORS is open to all origins on every endpoint.
Why it costs points	Anyone with the URL can insert arbitrary recommendations that your demo will then display as system output. A rival or a curious judge could literally make your platform recommend nonsense live on stage.
The fix	Delete or disable the POST branch for the demo (reads are all the demo needs). If you keep writes, require an auth token and validate the body against a schema. Time: 10 minutes to disable.

3. Full Problem & Solution Overview - Criterion by Criterion

3.1 Scientific Credibility: 3 -> 9

S1. All sensor data is synthetic (sine waves + seeded random)	NOW
The mistake	scripts/seed.js generates 90 days x 26 districts x 6 metrics using Math.sin/Math.cos/rng(), then labels the rows with real source names: Sentinel-2, CHIRPS, MODIS, MONRE, Landsat-9. The UI does include some "Simulated" disclaimers - that honesty scaffolding is good - but source badges still imply real provenance.
Why it costs points	Your own pitch promise is "all data real, anything estimated is labeled." Right now that claim is false. One question - "show me the raw CHIRPS file this reading came from" - and it collapses.
The fix	Do not try to make everything real tonight; make the RIGHT things real. Priority order: (1) real CHIRPS rainfall for the hero districts (it is a free, easy download - a few lines of Python per district); (2) real published salinity records for the 2016/2020 events (MONRE bulletins and published papers have the numbers); (3) keep the rest simulated but label it consistently: rename source to "Simulated (calibrated to CHIRPS climatology)". A demo that is 20% real where it matters and 80% honestly-labeled simulation scores far higher than one that is 100% fake and vaguely labeled.

S2. District boundaries are rectangles with real names	NOW
The mistake	public/mekong-districts.json contains 26 hand-drawn rectangular polygons named after real districts (Ba Tri, Binh Dai, ...).
Why it costs points	Dr. Dat has looked at real Mekong Delta maps his entire career. Rectangular districts scream "fake" from the back of the room and taint everything else on screen.
The fix	Download real admin-level-2 boundaries (GADM or opendevelopmentmekong provide free GeoJSON for Vietnamese districts), simplify with mapshaper to keep the file small, and swap the file. The app already reads GeoJSON, so this is a data swap, not a code change. Time: ~1 hour.

S3. Ghost jargon: "LSTM salinity", "model_version v2.4.1", "AI Decision Engine"	NOW
The mistake	Labels reference models that do not exist anywhere in the codebase. The actual logic is a transparent weighted scoring rule.
Why it costs points	Fake sophistication is a trap you set for yourself: the first "what architecture is the LSTM?" question has no answer. Meanwhile the rule-based system you DO have is defensible and explainable - a strength you are hiding.
The fix	Rename honestly: "Rule-based decision engine v1 (agronomic thresholds); ML forecast on roadmap." Prof. Vu spent a career on decision-support systems - he will respect a transparent, explainable rule engine far more than a fictional neural network. Time: 30 minutes of find-and-replace.

3.2 Technical Depth: 4 -> 7 now, 10 on roadmap

T1. Predictions are random numbers with fake confidence intervals	BEFORE DEMO
The mistake	The predictions table is filled with rng() values; uncertainty_low/high are just value x 0.8 / x 1.2.
Why it costs points	Random predictions cannot survive the chair's favorite question: "how much better is your forecast than a naive baseline?"
The fix	Replace with a real, simple, honest index for the hero districts: water-stress score = weighted(rainfall deficit vs. 30-yr norm, trend of last 30 days, salinity level vs. crop threshold). Computed from the real CHIRPS slice in S1, this becomes a genuine (if simple) model you can explain line by line. Then add the baseline: plot your index vs. a naive "last-value" rule on the back-test chart. Beating a baseline with a simple transparent model = real technical depth. ROADMAP slide: gradient-boosted / LSTM forecast trained on 20 years of CHIRPS+MONRE as v2.

T2. The same crop model is implemented twice with different weights	NOW
The mistake	api/simulator.js scores 5 factors (weights .15/.20/.25/.20/.20); CropSuitability.tsx scores 4 factors (.25/.30/.30/.15). The two pages can recommend different crops for the same scenario.
Why it costs points	If a judge opens the Simulator and the Crop Suitability page side by side and gets contradicting winners, your "decision engine" looks like a random-answer machine. This is the most dangerous LIVE-DEMO bug in the codebase.
The fix	Extract one scoring function into a shared module (src/lib/cropModel.ts), import it in both places, and source the weights from one constant with a code comment citing the rationale (e.g., FAO salinity-tolerance tables for the salinityMax values you already have). Time: ~45 minutes, and it upgrades Engineering Quality simultaneously.

T3. Missing the water-sustainability guardrail (the Chopra defense)	NOW
The mistake	Crop ranking optimizes profit under constraints but never asks whether the recommended crop's water use exceeds projected availability. Shrimp aquaculture (waterUse 15,000) can win in a drought scenario.
Why it costs points	Assoc. Prof. Chopra (sustainable urban systems, CityU HK) will ask the killer question: "does your system recommend crops that make the water crisis worse?" Right now the honest answer is yes.
The fix	Add one rule to the shared crop model: if crop.waterUse exceeds the scenario's water-availability budget, cap or flag the crop ("Not recommended: exceeds sustainable water budget"). Show the flag in the UI. Five lines of code; converts your key innovation from pure profit-seeking into climate-aligned economics and directly feeds the 30% impact score.

3.3 Engineering Quality: 5 -> 8

E1. Zero tests, template README, package named "placeholder-model-2"	BEFORE DEMO
The mistake	No test files exist. README.md is the unmodified Vite template. package.json name is a placeholder.
Why it costs points	Small signals, big inference: judges skim repos and infer care. A placeholder name says "generated, not owned."
The fix	NOW: rename the package, write a real README (what it is, how to run, data sources, honest data statement, architecture sketch). BEFORE DEMO: add 3-5 unit tests for the ONE thing that must not be wrong - the shared crop model (given drought scenario, salt-tolerant crop ranks first; given water budget X, over-budget crops are flagged). A tiny test suite on the core model reads as engineering maturity.

E2. Third-party auth proxy dependency (designarena.ai)	NOW
The mistake	Google login is routed through an external proxy tied to the tool that generated the code; it will fail on your own deployment.
Why it costs points	A login button that errors in front of judges is a self-inflicted wound, and auth adds zero value to a decision-support demo.
The fix	Remove Google auth from the demo path entirely. Keep the role-switcher as an explicit "Demo mode: choose your stakeholder view" - which is actually a better demo device than a login screen. Time: 20 minutes.

3.4 Innovation / Creativity: 6 -> 9

I1. The key innovation is buried	BEFORE DEMO
The mistake	The genuinely novel element - crop-switch economics (suitability x market price - switching cost, where "stay" can win) - is one page among nineteen, and switching costs are not yet in the profit math at all (profit = baseProfit x suitability x salinity haircut).
Why it costs points	Creativity judges score the sharpest original idea they can remember, not the feature count. Right now the memorable idea is diluted and half-implemented.
The fix	Add switching cost per crop (real establishment-cost figures exist in Vietnamese agricultural extension tables; label estimates as estimates) and a three-line scenario chart: switch now / switch later / stay, in currency. Then LEAD the pitch with it: "Every dashboard says the climate is changing. Ours tells you what to plant instead, what switching costs, and when staying is smarter." That sentence is your creativity score.

3.5 Presentation Quality: 6 -> 9

P1. 19 pages, no story	BEFORE DEMO
The mistake	Navigation exposes Mission Control, Live Operations, satellite passes, research dashboards, docs, architecture, deployment... The organizers explicitly instructed a LEAN MVP with extras on a roadmap.
Why it costs points	In 5 minutes you can show ~4 screens well. Nineteen options guarantee a wandering demo, and sprawl reads as "AI-generated feature soup" - a red flag this panel has seen many times this week.
The fix	Hide (do not delete) everything except: (1) Risk map, (2) District decision panel, (3) Crop suitability + switch economics, (4) Historical replay/back-test, (5) Impact. Demo script stolen from Project A's deck arc: Problem -> one farmer's journey -> the decision -> the proof (back-test) -> impact -> roadmap. Rehearse to 4 minutes 30 seconds.

P2. Impact claims phrased as achievements	NOW
The mistake	UI and companion deck use achievement language ("10,000+ farmers protected", "3x income") for a system that has never been deployed.
Why it costs points	Mr. David Falcon (Director of ESG, VinFast) verifies impact claims for a living. Unverifiable achievement claims are his easiest kill, and they poison your real numbers too.
The fix	Reframe every impact number as designed capacity or target with visible arithmetic: "Covers 26 districts / ~X hectares; at a conservative 10% adoption and Y% avoided loss (source), estimated Z per season." Small transparent numbers beat big hollow ones with this panel, every time.

3.6 Business Potential: 5 -> 8

B1. No first customer, no willingness-to-pay story	BEFORE DEMO
The mistake	Role dashboards imply the business model but nothing names who signs first or why they pay.
Why it costs points	Two judges run university entrepreneurship centers (Dr. Paul Wang, HKU; Assoc. Prof. Carrie Ling, HKUST). "Who is your first customer?" is guaranteed. "Cooperatives in general" is not an answer.
The fix	One slide: first customer = a named TYPE with a number ("rice-shrimp cooperatives in Ben Tre and Soc Trang provinces, ~N co-ops covering M hectares"); who pays = co-op pays per-hectare advisory fee or insurer/agribusiness sponsors, farmers free; why now = post-2020 salinity losses made prevention cheaper than compensation. If you have spoken to ANY cooperative or agronomist, say so; if not, name it as step one - do not fake traction.

3.7 Impact & Scalability: 4-5 -> 8

M1. Impact analytics computed over fabricated inputs	BEFORE DEMO
The mistake	The Impact page aggregates seeded fake values, so every headline metric is theater.
Why it costs points	This feeds the official 30% impact criterion. Fake inputs cap the score no matter how nice the charts are.
The fix	Recompute the headline impact from the honest back-test (C2/T1): "In the 2016 replay, our index crossed the warning threshold N weeks before documented crop losses; acting at that point protects an estimated X% of the winter-spring rice value across the affected districts (arithmetic shown)." One real, defended number outweighs six invented ones.

M2. Scalability is asserted, not designed	ROADMAP
The mistake	Hardcoded district list, no ingestion pipeline, no schema/migrations in the repo; meanwhile the deck claims seven nations.
Why it costs points	The gap between claim and code is where scalability scores die.
The fix	ROADMAP slide with an honest ladder: v1 hero provinces (now) -> full delta via automated CHIRPS ingestion (monthly cron) -> cross-border by swapping the data layer (the engine is data-source-agnostic - your genuinely true portability claim). Cut "INFINITE IMPACT" language entirely.

4. Judge-by-Judge Defense Map

Judge	Background	Most likely attack	Your prepared answer / fix
Prof. Duong Nguyen Vu (Chair)	Decision-support systems for air traffic (NTU/ATMRI, EUROCONTROL); VinUni CSO for AI research	"What exactly does the model predict, and is it better than a naive baseline? How do you handle uncertainty?"	Baseline line on the back-test chart (T1); revenue shown as ranges; say the word "uncertainty" explicitly; transparent rule engine, not fictional ML (S3).
Assoc. Prof. Nguyen Trung Thanh	Deputy Head, Data Science & AI department	"Walk me through the back-test methodology. Train/test split? Leakage?"	Honest framing before he asks: "retrospective validation on 2016 with a threshold rule frozen in advance; walk-forward validation is our next step" (C2 Option A).
Dr. Nguyen Duy Dat	Head of Environmental Technology, HCMC Univ. of Technology; knows the Mekong intimately	"Which salinity thresholds? Where is this data from? Those boundaries look wrong."	Real district geometry (S2); real 2016/2020 salinity records with sources on screen; humble close: "we would value expert guidance on Mekong-specific factors."
Assoc. Prof. Shauhrat S. Chopra	School of Energy & Environment, CityU HK; sustainability & resilience	"Does maximizing profit push farmers toward MORE water-intensive crops?"	Show the water-budget guardrail live (T3): pick the drought scenario, point at the flagged over-budget crop. This one moment can win his score.
Dr. Paul Wang	Director, Technology Entrepreneurship Centre, HKU	"Who pays? Who is customer #1? Why you?"	The one business slide (B1): named customer type, per-hectare fee, insurer sponsorship, why-now.
Assoc. Prof. Carrie Ling	Director, Entrepreneurship Center, HKUST; systems design	"Has a real cooperative manager used this? Could they?"	Demo through the cooperative persona start to finish (P1); honest user-validation status; roadmap step one = pilot with a co-op.
Mr. David Falcon	Director of ESG, VinFast	"How is 'losses prevented' computed? Can it be verified?"	Arithmetic on screen (P2/M1); bonus line: "metrics designed to be auditable, the way ESG reporting requires."

5. Priority Action Plan (ordered - do top to bottom)

#	Action	Est. time	Scores it moves
1	Rotate Supabase keys; strip secrets from vercel.json; disable open POST endpoint (C1, C3)	25 min	Deployment, Feasibility 40%
2	Delete fabricated 0.82 accuracy / $120M avoided figures; relabel or recompute (C2)	20 min - 4 h	Credibility, Q&A survival
3	Unify crop model into one shared module; add water-budget guardrail (T2, T3)	1 h	Engineering, Impact 30%, Chopra
4	Real district GeoJSON swap (S2)	1 h	Credibility, Dat
5	Real CHIRPS slice for hero districts + honest source relabeling (S1)	2-3 h	Credibility, Feasibility 40%
6	Real simple water-stress index + naive baseline on back-test chart (T1)	2-3 h	Technical depth, Vu & Thanh
7	Add switching costs + switch-now/later/stay chart; make it the pitch centerpiece (I1)	1.5 h	Creativity 15%
8	Cut demo nav to 5 pages; rehearse the 4:30 story (P1); reframe impact numbers with arithmetic (P2, M1)	1.5 h	Presentation 15%, Impact 30%

9	Business slide with named first customer (B1); remove auth, keep demo-mode role switcher (E2)	45 min	Business, Wang & Ling
10	Real README + 3-5 unit tests on the crop model (E1); honest roadmap slide (M2, T1-v2)	1 h	Engineering, Scalability

If time runs out: items 1-4 are non-negotiable. Item 5 can shrink to ONE hero district. Item 6 can shrink to the 2016 event only. Items 7-10 degrade gracefully into spoken answers instead of built features.

6. What 10/10 Actually Looks Like on Stage

You open with one farmer in one district facing one decision. You show the risk map (real boundaries), drill into the district (real rainfall history for it), and the engine recommends a switch - or recommends staying, with the switching cost that justifies it. You click the drought scenario and one crop gets flagged: "exceeds sustainable water budget" - the system refuses to solve a climate problem by creating a water problem. Then the proof: the 2016 replay, your index crossing the threshold weeks before the documented losses, next to a naive baseline that missed it. Then impact, as arithmetic anyone can check. Then one business slide, one roadmap slide, and you stop at 4:30 with the sentence the judges will repeat to each other: **"the team that tells you what to plant, what switching costs, and proved it on the real 2016 crisis - honestly."** Every claim on screen survives every question in the room. That is a 10.

7. Open Questions for the Team

1. **Back-test status:** is anything real behind the Historical Replay yet, or is it 100% the seeded values? This decides between Option A (compute it) and Option B (relabel it) in C2 - answer before starting item 2.

2. **Hours remaining and division of labor:** how many working hours do you two realistically have before submission (deadline 04 July, 08:00 GMT+7), and can you work in parallel (one on data, one on code)? The plan above assumes ~8-10 combined hours.

3. **Deployment control:** do you own the Supabase project and the deployment (can you rotate keys and redeploy), or was it provisioned by the generation tool? If you cannot rotate the key, migrate the seed to a fresh Supabase project you own - ~30 minutes.

4. **Language/pivot check:** both artifacts are now Mekong-first, not southern-India-first as originally planned. Intentional? For this venue and this panel it is the right call - but your proposal document must match what you demo.

Prepared as a simulated judge's review for team preparation - Asian Hackathon for Green Future 2026. All findings are from direct inspection of the submitted codebase and slide deck; judge profiles from public announcements.